

Comparison Between Run 1 and Run 2

I executed the full evaluation twice using random 80/20 splits without reproducibility. Because the splits differ each time, the mean accuracy and standard deviation naturally vary. Below is a direct comparison of two runs.

Zoo Dataset

Run 1:

Exact Bayes: mean 0.7048, std 0.0595

Naive Bayes: mean 0.9238, std 0.0381

Linear Regression: mean 0.7714, std 0.0700

Run 2:

Exact Bayes: mean 0.6619, std 0.0497

Naive Bayes: mean 0.9238, std 0.0530

Linear Regression: mean 0.7524, std 0.0819

Weather Dataset

Run 1:

Exact Bayes: mean 0.3167, std 0.1167

Naive Bayes: mean 0.5000, std 0.1054

Linear Regression: mean 0.6167, std 0.1675

Run 2:

Exact Bayes: mean 0.2667, std 0.1528

Naive Bayes: mean 0.4833, std 0.1893

Linear Regression: mean 0.5833, std 0.2141

Overall Comparison

Naive Bayes is the most stable and best performer on Zoo.

Exact Bayes fluctuates the most across runs.

Linear Regression shows moderate variance on Zoo but higher variance on Weather.